ii) Spread plate method

- Spread plate method is an easy and direct method of isolating a pure culture.
- In this technique a specified amount of diluted inoculum (0.1ml or less) of microbial culture is seeded on agar plate.
- After inoculation of the sample on the agar medium, the inoculum is evenly spread on the surface with the help of a sterile glass L rod (a bent glass rod)
- Microorganisms are evenly distributed in the entire surface of agar.
- The dispersed microorganisms develop into isolated colonies.
- In this method, the plates are incubated at a specified temperature for a given period of time.
- After incubation the plates are observed for the growth of discrete colonies.
- The number of colonies are equal to the number of viable organism. This method can be used to count the microbial population (Figure 5.11).

iii) Streak plate technique

- The streak plate technique is one of the most commonly used methods for isolating pure culture of bacteria.

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- In this method, a loopful of inoculum from a sample is taken and it is streaked across the surface of the sterile solid medium.
- Different streaking patterns can be used to separate individual bacterial cell on the agar surface.
- After the first sector is streaked the inoculated loop is sterilized and inoculum for the second sector is obtained from the first sector.
- Similar process is followed for streaking the further areas in the sectors.
- Since the inoculum is serially diluted during streaking patterns the dilution gradient is established across the surface of the medium.
- After streaking, plates are incubated at a specific temperature for a given period of time.

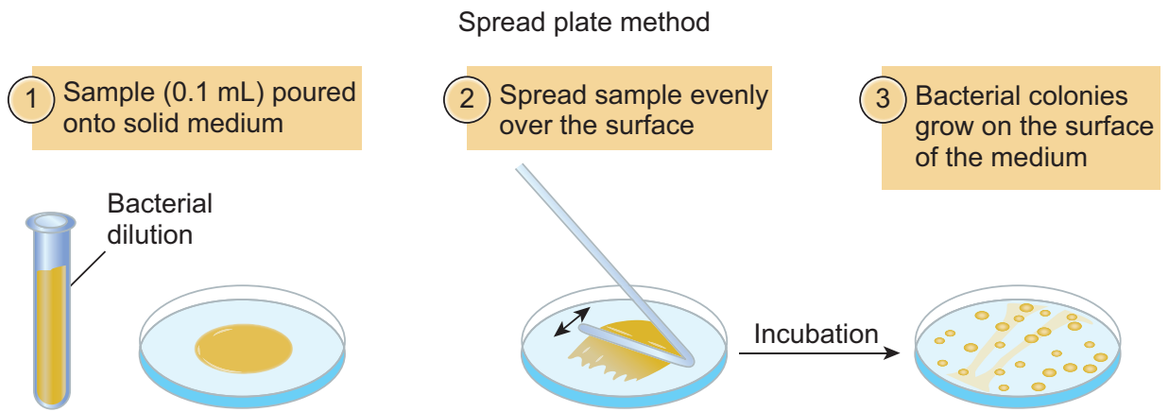


Figure 5.11: Spread plate method

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